

SDI Intelligence server: who does what

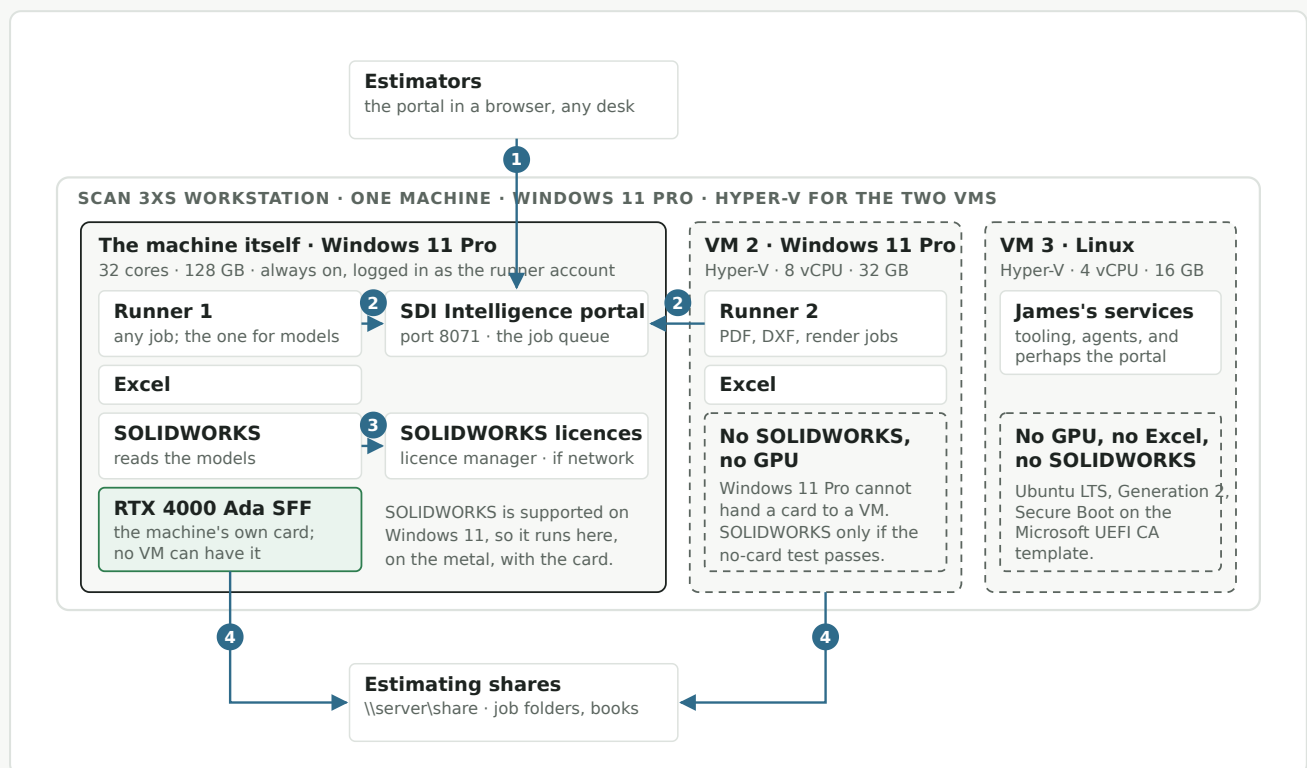
Revised 5 October 2026 · the machine runs Windows 11 Pro itself, with a Windows 11 VM and a Linux VM under Hyper-V; the earlier plan had a Windows Server 2025 host with a Windows 11 virtual machine for SOLIDWORKS

Jack builds the machine and hands over a Windows 11 Pro workstation that behaves like a new laptop on the network, with the graphics card in it, plus two virtual machines on it: one Windows 11 Pro, one Linux. James takes it from there, exactly as he would set up a laptop. The handover checks in the middle are the line between the two. The first section shows how the parts fit together, what changed from the Server plan, and what to watch for.

Scan 3XS · Threadripper PRO 9975WX (32 cores) · 128 GB ECC · NVIDIA RTX 4000 Ada SFF 20 GB · 2 × 2 TB NVMe · Windows 11 Pro on the machine itself · SOLIDWORKS, Excel, runner 1 and the portal on it · Hyper-V: VM 2 Windows 11 Pro (runner 2), VM 3 Linux (services)

How it fits together

One machine, three operating systems. SOLIDWORKS, Excel, the first runner and the portal run on Windows 11 Pro directly, and the graphics card is the machine's own. Under Hyper-V on the same box: a Windows 11 Pro VM for a second runner, and a Linux VM for the services James names. Neither VM has a graphics card.



1. Estimators start a job on the portal. It joins the queue on the machine.

2. Each runner, when it is free, takes the oldest job waiting. One job at a time each; runner 1 is on the machine itself, runner 2 in the Windows VM. The Linux VM runs no estimates: it has no Excel and no SOLIDWORKS.
3. SOLIDWORKS takes a licence while it is open. The runner keeps it open, so the machine holds one seat all day. With a standalone licence the seat simply lives on this machine.
4. The runners read and write the job folders and books on the estimating shares, which stay on the file server.

WHAT CHANGED FROM THE SERVER 2025 PLAN

SERVER 2025 PLAN	WINDOWS 11 PRO PLAN	EFFECT
Windows Server 2025 host, Windows 11 VM for SOLIDWORKS, the RTX 4000 passed through to that VM (DDA)	Windows 11 Pro on the metal; SOLIDWORKS, Excel, the runner and the portal all on it; the card is simply the machine's	One operating system instead of two, no pass-through to set up, no VM licence for the first runner. SOLIDWORKS is supported on Windows 11 and was never supported on the Server host.
Portal service on the host; runner in the VM	Both on the same machine; the runner points at <code>localhost</code> or the machine's own name	Nothing in the engine changes. A restart takes both down together, which it did in effect before.
Windows Server licence, plus Windows 11 and Office for each VM	Windows 11 Pro comes with the workstation; Office for the machine; one Windows 11 Pro and one Office licence for VM 2; nothing for the Linux VM	No Server licence, no client access licences. Cheaper and simpler.
Hyper-V only if a second VM was ever wanted	Hyper-V on from day one, for the Windows VM and the Linux VM	Windows 11 Pro includes Hyper-V and runs Linux guests well (Ubuntu ships the integration drivers). The VMs need an External virtual switch so they sit on the office network with fixed addresses.
A second card could be passed through to VM 2	It cannot: Windows 11 Pro has no DDA. A second SOLIDWORKS runner with a card means a second physical machine	VM 2 is for PDF, DXF and render jobs. The no-card test below decides whether it could also run SOLIDWORKS in software graphics.

SERVER 2025 PLAN	WINDOWS 11 PRO PLAN	EFFECT
SOLIDWORKS 2026 needed for Hyper-V 2025 support	Any version Dassault supports on Windows 11; 2026 still recommended	The Hyper-V constraint only applies if SOLIDWORKS is ever put in VM 2.
Two admin sessions at once (Server)	One interactive session (Windows 11). See the gotchas	Remote access has a rule to follow so it does not disturb the runner.

WHAT CAN AND CANNOT BE SHARED

The graphics card: the machine's own, and nobody else's.

On Windows 11 Pro the RTX 4000 Ada drives the machine's own desktop and SOLIDWORKS directly, with NVIDIA's RTX workstation driver. It cannot be handed to a virtual machine: Discrete Device Assignment is a Windows Server feature, and the partitioning Windows 11 does have (GPU-P) is unsupported by NVIDIA and Dassault for this card and breaks on driver updates. So SOLIDWORKS with a card runs on the metal, which is where this plan puts it.

The estimates: shared by the queue.

This is where "whoever is free takes the next one" happens. A second runner VM means two estimates at once; each takes the oldest job waiting when it finishes the last.

SOLIDWORKS: the licences are shared, the program is not.

Each machine or VM that uses SOLIDWORKS needs its own installation. Now that SOLIDWORKS runs on a physical machine, a standalone licence works without the reactivation risk a VM carried. A network (SolidNetWork) licence is still the right choice if designers' PCs should share the pool; the licence manager can run on this machine (it is supported on Windows 11) or on the file server.

Test before planning a second SOLIDWORKS runner.

The model extract may not need the graphics card at all: its geometry work runs on the processor, and SOLIDWORKS falls back to software graphics without a card. Run the extractor on job 12120 in VM 2 (no card) and compare the result and the time with the machine itself. If they match, VM 2 can run SOLIDWORKS too, with a second licence and no hardware.

The estimating shares stay on the file server.

Windows 11 Pro allows twenty inbound connections to its own shares. The portal is a web service (no limit of that kind) and the runners read the job folders over the network, so nothing here needs this machine to be a file server, and it should not be made one.

IF MORE SOLIDWORKS CAPACITY IS EVER NEEDED

OPTION	WHAT IT GIVES	CATCH
SOLIDWORKS in VM 2 without a card	Two model-capable runners on one box	Only if the no-card test passes. A second SOLIDWORKS licence. Software graphics are slow for a person but may be fine for an unattended extract.
A second physical workstation as runner 2	A card and SOLIDWORKS on each runner	A second machine, Windows 11 Pro, Office and SOLIDWORKS licences. Supported and simple; nothing to share.
Reinstall this machine as Windows Server 2025 with VMs	The original plan: DDA pass-through, two admin sessions	Server licence, SOLIDWORKS back inside a VM, and a rebuild. Only worth it if several model runners must live on one box.
Share this card with a VM through GPU-P	Host and VM share one card	Unsupported by Microsoft for workstation cards, NVIDIA and Dassault; breaks on driver updates. Not for this machine.

Gotchas: a workstation OS doing a server's job

None of these stops the plan. Each one is a setting or a habit, and each is on a list below with an owner.

1. **One interactive session.** Windows 11 allows one desktop at a time. If James connects by Remote Desktop as himself, the runner account's desktop is pushed off the console, and Excel and SOLIDWORKS automation can stall or fail mid-estimate. Rule: connect *as the runner account* (same session), and leave by handing the session back to the console with `tscon` (a one-line script on the desktop), never by closing the window, which leaves the session disconnected and lockable. Or use a console-sharing tool (VNC-style) rather than RDP. Never remote in while a job is running unless it is to watch it.
2. **Sleep, lock and fast startup are on by default.** Power plan to never sleep, never turn off the display to the extent policy allows, hibernate and fast startup off, screensaver and lock off for the runner account. A domain lock policy needs an exemption for this machine. A locked or sleeping desktop is a stopped runner.
3. **Windows Update is more insistent than on Server.** Feature updates arrive yearly and restarts are pushed harder. Windows 11 Pro takes Group Policy: set active hours, schedule restarts out of hours, defer feature updates, and tell James before any restart. A restart mid-run loses that estimate.
4. **No graphics card for any VM.** SOLIDWORKS with a card runs only on the metal. VM 2 is for jobs without models until the no-card test says otherwise.
5. **Auto-logon of a domain account.** Windows 11 hides the automatic sign-in setting when Windows Hello is required; use Sysinternals Autologon for the runner account and turn Hello off for it. The password is stored on the machine, so it is a dedicated, least-privilege account.
6. **Hyper-V changes the host, and it is on from day one.** With Hyper-V on, Windows itself runs as a partition, with a small cost to SOLIDWORKS and Excel on the metal; the two VMs need it, so that cost is accepted. Give the VMs fixed shares of the machine (Dynamic Memory off for the Windows VM) and keep at least 64 GB and 16 cores for the metal, where the models are read. Memory integrity (VBS), on by default, costs a few percent of processor speed; turning it off is a security trade-off for Jack to decide.
7. **The VMs must be on the office network, not behind the host.** Hyper-V's Default Switch is a private NAT whose addresses change on every boot. Create an External virtual switch on the machine's network port so both VMs get fixed addresses and DNS names, and the runners, the portal and Remote Desktop can reach them by name.
8. **Linux under Hyper-V has two settings of its own.** A Generation 2 Linux VM boots with Secure Boot only on the "Microsoft UEFI Certificate Authority" template (the Windows template refuses it); and there is no Enhanced Session desktop unless xrdp is set up, so it is administered over SSH. Ubuntu LTS carries the Hyper-V integration drivers in its kernel, so nothing extra is installed.
9. **No graphics card in the Linux VM either,** for the same reason as the Windows VM. Anything there that wants a GPU (a local model, say) runs on the processor.
10. **No Windows Server Backup.** Use an imaging tool (Veeam Agent, Macrium or the company standard) for the machine and the VM, and test one restore before go-live.

11. **Pro or Pro for Workstations.** Scan may ship Windows 11 Pro for Workstations with a Threadripper PRO. Either works. Plain Pro covers one socket, 32 cores and 128 GB; Pro for Workstations adds ReFS and more than two sockets or 2 TB, none of which this needs.
12. **Power after a cut.** BIOS set to power on after AC loss, the machine on the UPS, and the runner logging on by itself on boot, so an overnight cut does not leave the queue dead until someone notices.

JACK Infrastructure: the machine

Everything below the applications: hardware, operating system, network, accounts, backup. In build order.

THE MACHINE (WINDOWS 11 PRO)

1. **BIOS:** latest firmware; power on after AC loss; enable virtualisation (SVM) and IOMMU, which Hyper-V needs for the two VMs.
2. **Storage:** set up the two NVMe drives (mirror, or OS on one and data and VM on the other — Jack's call) and install Windows 11 Pro with current drivers, including NVIDIA's RTX workstation driver for the 4000 Ada.
3. **Network and domain:** fixed IP, DNS name, join the domain.
4. **Runner account:** a dedicated domain user that logs on automatically at boot (Sysinternals Autologon; Windows Hello off for it); read/write to the estimating shares by UNC path (`\\server\share`), not K:. The portal service runs under this account or its own service account with the same share rights.
5. **Power and lock policy:** no sleep, no hibernate, fast startup off, no screen lock, no screensaver — SOLIDWORKS and Excel need a live desktop session. Exempt the machine from any domain lock policy.
6. **Windows Update policy:** active hours, restarts scheduled out of hours and announced, feature updates deferred.
7. **Firewall:** allow the portal port (`8071`) from the office network.
8. **Remote access for James:** Remote Desktop as administrator, with the one-session rule from the gotchas; the admin share, which the laptop's `push-to-server.ps1` copies through.
9. **SOLIDWORKS licence manager** (SolidNetWork), only if the licence is a network one: install it on this machine (or the file server) with the reseller's serial and open ports `25734` and `25735` to designers' PCs if they share the pool. It must be at least as new as the SOLIDWORKS install.

HYPER-V AND THE TWO VMS

10. **Turn on Hyper-V** (a Windows 11 Pro feature) and create an **External virtual switch** on the machine's network port, so the VMs are on the office network with their own fixed addresses.
11. **VM 2, Windows 11 Pro:** Generation 2 with Secure Boot and a virtual TPM (Windows 11 requires both); 8 virtual CPUs, 32 GB memory with Dynamic Memory **off**, a 200 GB+ virtual disk on NVMe. Fixed IP and DNS name, joined to the domain, its own runner account with auto-logon, no sleep or lock, fully updated. **No graphics card and no**

SOLIDWORKS. Its runner goes live only after James's routing change (see his list), so jobs with models always go to the machine itself.

12. **VM 3, Linux:** Generation 2 with Secure Boot on the **Microsoft UEFI Certificate Authority** template (no virtual TPM needed); 4 virtual CPUs, 16 GB memory, a 100 GB+ virtual disk. Ubuntu LTS server, fixed IP and DNS name, SSH open to James, an account for him with sudo. With both VMs at full size the machine itself keeps 20 cores and 80 GB.
13. **Both VMs start automatically** when the machine boots, after the host has logged the runner account in.

RUNNING IT

14. **Licences:** Windows 11 Pro comes with the workstation; one Office licence for it. VM 2 needs its own Windows 11 Pro and Office licences. The Linux VM needs none. No Server licence and no client access licences.
15. **Backups:** an imaging tool for the machine and an export of each VM, with one restore tested before go-live.
16. **Antivirus:** apply the exclusions James supplies (SOLIDWORKS temp and the output folders).
17. **Power:** on the UPS, BIOS set to start again after a power cut, the runner logging on by itself.

Handover checks

Jack confirms these, then the machine is James's:

- In Device Manager the graphics card shows as **NVIDIA RTX 4000 Ada** with the NVIDIA driver, not "Microsoft Basic Display".
- After a full restart, the machine logs in as the runner account with nobody touching it, and is still logged in and unlocked an hour later; both VMs are running, and each answers on its own fixed address by name.
- From James's laptop, SSH into the Linux VM works by name, and Remote Desktop into the Windows VM works as its runner account.
- Signed in as the runner account, it can open and save a file in the estimating folders by UNC path.
- From a desk, the machine answers on port 8071 (once James's service is running, this is just a browser test).
- If the licence is a network one: from a designer's PC, the licence manager answers on port 25734.
- James can Remote Desktop in as an administrator, and the `tscon` hand-back script is on the runner's desktop.

JAMES

Applications: the laptop premise

Everything that makes it an SDI Intelligence machine, installed the same way as on the laptop — and now on one machine, as on the laptop.

ON THE MACHINE (RUNS THE ESTIMATES AND THE PORTAL)

1. **SOLIDWORKS:** install and activate (standalone), or point it at the licence manager (network). Open one model to confirm the graphics card is in use.
2. **Office / Excel:** install and activate; open one estimate book.
3. **PDM / Design Vault client,** if models come from the vault.
4. **Python and Git,** clone the repository, create the engine's `.venv`.
5. **The `.env` file,** written by hand for this machine — never copied from the laptop.
6. **The portal:** the backend's own `.venv` and `.env`; install and start the SDIIntelligence service; check the portal loads from a desk.
7. **The runner:** scheduled task at logon (`restart-runner.ps1`) pointing at the portal on this machine.
8. **Hand Jack the antivirus exclusion paths.**

PROVING IT

9. **Run one known job** (12645 or 12173) and compare the book and the run time with the laptop's.
10. **Restart the machine** and confirm the portal and the runner both come back without anyone logging in.
11. **Remote in and out once** as the runner account while a short job runs, hand the session back with `tscon`, and confirm the job finished.

ON THE LINUX VM

12. **What it runs is James's to name:** tooling, agents, and any service that needs no Windows. The portal could live here instead of on the metal (it is a Python service and runs the same on Linux); if it does, the runners point at the Linux VM's name, it runs under systemd, and it needs no access to the estimating shares, which only the runners read and write.
13. **Nothing that drives Excel or SOLIDWORKS** goes here: those are Windows programs and the runners stay on Windows.

BEFORE RUNNER 2 GOES LIVE

14. **Routing change in the engine:** today whichever runner is free takes the next job, so a job with SOLIDWORKS models could land on VM 2 and be costed from the PDFs alone. Each runner will say whether it has SOLIDWORKS, and jobs with models will wait for one that does.
15. **The graphics test:** run the SOLIDWORKS extractor on 12120 in VM 2 with no card and compare the extract and the time with the machine itself. This decides whether VM 2 can ever run SOLIDWORKS.

Agree together before the build week

Rule of thumb: if it would be the same on any Windows machine, it's Jack's. If it only exists because this is SDI Intelligence, it's James's.

QUESTION	WHO LEADS	WHY IT MATTERS
SOLIDWORKS licence type: standalone on this machine, or network (SolidNetWork) shared with designers; cover for unattended automated use	James, with the reseller	On a physical machine a standalone licence is simple and does not need reactivating. A network licence puts the seats in a pool, first come, first served. Either way the runner keeps SOLIDWORKS open and holds one seat all day.
SOLIDWORKS version: 2026 recommended	James, with the reseller	Any version Dassault supports on Windows 11 runs on the metal. The Hyper-V constraint (2026 for Hyper-V 2025) only matters if SOLIDWORKS is ever put in VM 2.
Windows 11 Pro or Pro for Workstations	Jack, with Scan	Either runs everything here. Take whichever Scan supplies; nothing in the engine needs the Workstations edition.
When runner 2 goes live on the Windows VM	James	Two runners means two estimates at once. The VM is built with the machine; its runner waits for the routing change and its own Windows 11 and Office licences.
What the Linux VM is for, and whether the portal moves to it	James	Decides its size, what Jack opens on the firewall, and which machine name the runners point at. Nothing about SOLIDWORKS or Excel changes either way.
Fixed shares of the machine for the two VMs	Jack and James	The models are read on the metal, so it keeps most of the cores and memory: 20 cores and 80 GB with the VMs as sized here.
A second SOLIDWORKS runner	James, after the graphics test	Either VM 2 without a card (if the test passes) or a second physical workstation. A second card in this machine helps no VM on Windows 11 Pro.

QUESTION	WHO LEADS	WHY IT MATTERS
Machine name and IP address	Jack	The runner, the portal links and the service settings all name it.
Who is told before Windows updates or restarts, and the remote-access rule	Jack and James	A restart or a careless Remote Desktop session mid-run loses that estimate.